# MATH 190 (Section 101): Calculus Survey

https://pabloocal.github.io/Teaching/2026-2027-WT1-MATH190-101/

2026-2027 Winter Term 1

## Instructor

**Name:** Pablo S. Ocal
**Office:** MATX 1208
**Contact:** https://pabloocal.github.io/Contact/
**Lectures:** MWF 14:00-14:50

**Office hours:**
W 15:00-16:30

**Classroom:** IRC G3

## Prerequisites

- The prerequisites required for this course are covered in Pre-calculus 12.
- You are expected to be familiar with polynomials and rational functions, including manipulating them and knowing their graphs.
- You are expected to be somewhat familiar with trigonometric functions, exponential functions, and logarithm functions, including manipulating them and knowing their graphs.

## Textbooks

- *Single Variable Calculus (2nd edition)* by Jonathan D. Rogawski.
- *CLP-1 Differential Calculus* by Joel Feldman, Andrew Rechnitzer, and Elyse Yeager.
- *CLP-2 Integral Calculus* by Joel Feldman, Andrew Rechnitzer, and Elyse Yeager.

## Quizzes

There will be five in-class quizzes throughout the course. Quizzes will consist of two exercises. During quizzes you must work on your own, you are not allowed to work in groups. Quizzes will be graded and promptly returned to you via Canvas. Your score on each *exercise* will be based on completeness, correctness, and on your work shown, according to the following scale. **Missed quizzes will be given a grade of 0.**

| | |
|---|---|
| Blank or no submission | **0 points** |
| Reasonable effort | **6 points** |
| Minor mistakes | **10 points** |
| No mistakes | **12 points** |

## Laboratories

There will be ten laboratories and one laboratory review throughout the course. Laboratory assignments will consist of two exercises. During laboratories you are encouraged to work in groups, but submissions are individual, and the work you submit must be your own. Assignments will be graded and promptly returned to you via Canvas. Your score on each *exercise* will be based on completeness, correctness, and on your work shown, according to the following scale. **Late assignments will not be accepted. Submissions after the due date will not be accepted.**

| | |
|---|---|
| Blank or no submission | **0 points** |
| Wanting effort | **1 point** |
| Reasonable effort | **2 points** |

## Homework

There will be eight homework assignments throughout the course. Homework assignments will consist of five exercises. You are encouraged to discuss homework assignments in groups, but submissions are individual, and the work you submit must be your own. Assignments will be graded and promptly returned to you via Canvas. Your score on each *exercise* will be based on completeness, correctness, and on your work shown, according to the following scale. **Late assignments will not be accepted. Submissions after the due date will not be accepted.**

| | |
|---|---|
| Blank or no submission | **0 points** |
| Wanting effort | **1 point** |
| Reasonable effort | **2 points** |

## Exams

There will be one midterm exam and one final exam. During the exams, you may not use notes, calculators, cell phones, or anything other than pen. Your score on each exam will be based on completeness, correctness, and on your work shown, according to the rubric. **Missed midterm exams will be given a grade of 0. You must take the final exam.** Please bring a photo ID to every exam. The exams are scheduled for the following dates:

- Midterm: November 2.
- Final: TBD.

## Grading

Your final grade in this class will be computed as follows.

| | **Grade** |
|---|---|
| Quizzes | **15%** |
| Laboratories | **5%** |
| Homework | **10%** |
| Midterm | **20%** |
| Final | **50%** |

## Accommodations

If you have a condition that requires accommodation, please contact the Centre for Accessibility as soon as possible. They will determine with you what accommodations are appropriate and communicate them to the instructor. This service is confidential. **If you require accommodation, you will take the exams at the facilities of the Centre for Accessibility. It is your responsibility to make sure you have contacted the Centre for Accessibility and that they are expecting you. It is your responsibility to make sure the Centre for Accessibility contacts the instructor in time to provide the exams.**

## Academic Integrity

This course is subject to UBC's Policies and Regulations. The work in this class is expected to follow UBC's Academic Integrity guidelines. All work submitted for credit must be the student's own and should reflect the student's own understanding of the material.

## Generative Artificial Intelligence

The following is the instructor's viewpoint at the time of writing and is subject to change. The instructor sees generative artificial intelligence as a tool akin to an abacus, a physical calculator, an online calculator, or a computer. **It is important to understand that generative artificial intelligence frequently makes mistakes and that we do not yet know how to reliably use it.** Ultimately, in the same way that someone who is not proficient at using an abacus should not be trusted with the output of their fiddling, we are not (yet) proficient at using generative artificial intelligence and we should be extremely careful when handling the output of our prompts.

As such, you are encouraged to use generative artificial intelligence to enhance your learning of the material covered in the course, and you are warned not to rely on generative artificial intelligence to think for you nor to solve the exercises for you. The recommended workflow to solve exercises in this course is as follows.

1. Invest 20 minutes in thinking about the exercise on your own, without aids.
2. If you seem to be making progress, allow yourself another 10 or 20 minutes of unaided work.
3. If you are not making progress, invest 20 minutes working with the coursebook or your class notes open, looking for similar exercises that could give you ideas on how to solve the exercise at hand.
4. If you are not making progress, use Wolfram|Alpha or Desmos to guess what the solution could be. Invest 20 minutes working towards that solution.
5. If you are not making progress, you may consider using generative artificial intelligence to guide you, but I would instead recommend you come to office hours. They happen every week.

You are allowed to use generative artificial intelligence exactly when you are allowed to use a calculator, but you are strongly encouraged to use generative artificial intelligence as the very last resort. Always keep in mind that when you submit work for credit, that work must be your own, and must reflect your own understanding of the material.